From Transient Thunderstorm Wind Fields to Structural Design Loads: Response-Translation Methods, Standard-Interface Boundaries, and a Bounded PETW Upstream-Benchmark Case Study

Internal PI review draft; scientific status not approved.

Abstract

Transient wind fields produced by thunderstorm downbursts differ from stationary atmospheric boundary layer (ABL) assumptions in vertical profile shape, pulse duration, turbulence nonstationarity and aerodynamic transfer. Within a mechanism-conditioned hazard-to-load architecture, this study places the gust-front factor (GFF) and the generalized gust-front factor (G-GFF) within a generalized mechanism-conditioned wind-force-response chain, distinguishes their denominators and aligned limiting behavior, and treats the ASCE 7-22 interface as a candidate reporting layer. GFF pairs a scalar response ratio with a height-dependent kinematic coefficient against a fully recomputed ASCE 7-22 baseline, expressing the mechanism-conditioned load as a multiplier on the baseline load. G-GFF uses the mechanism-wind static-mean response as denominator, forming a self-contained load expression decoupled from any specific code format. Both belong to the same response-translation family and, under aligned inputs, express the same target load effect; they must not be stacked as independent gains alongside the thunderstorm response spectrum (TRS) or equivalent static wind load (ESWL). This study also checks edition-specific bookkeeping against ASCE 7-16 as a historical reference. Within the standards and literature examined, the reviewed evidence does not support calibrated transient wind-load reliability factors, conservative defaults, or directly adoptable downburst design multipliers. A bounded PETW case study evaluates upstream reproducibility, definition sensitivity and time-semantic gates; its diagnostic evidence does not calibrate a hazard wind-speed quantity, structural response factor, or design multiplier.

1 Introduction

Classical structural wind load design theory rests on stationary atmospheric boundary layer (ABL) assumptions: mean wind speed profiles following logarithmic or power laws with height, stationary Gaussian turbulence processes, and gust loading factors defined within this framework [Zhou et al., 2002]. ASCE 7-22 defines the basic wind speed V as a nominal 3 s gust at 10 m height under Exposure C conditions [American Society of Civil Engineers, 2022]; the velocity pressure $q_z = 0.613\,K_z\,K_{zt}\,K_e\,V^2$ (SI units, Eq. 26.10-1.SI, printed p. 277) excludes the directionality factor K_d, which appears in downstream pressure or force equations. The exposure coefficient $K_z = 2.41(z/z_g)^{2/\alpha}$ is capped at 2.41 above z_g (printed pp. 275-278). This complete itemized baseline chain provides a mature design procedure for main wind force resisting systems (MWFRS) and components and cladding (C&C) under ABL conditions.

The wind-field characteristics of thunderstorm downbursts differ fundamentally from the ABL assumptions described above [Solari, 2020]. The radial outflow forms a non-monotonic nose-shaped vertical profile near the surface, and wind speed evolves as a finite-duration pulse process over time. The records compiled by Kwon and Kareem yield a mean duration of about

248 s and a minimum of about 57 s, but these statistics apply only to the event sample and definition adopted in that source [Kwon and Kareem, 2019] (Eqs. 1-4, PDF p. 2). The turbulence process exhibits nonstationary amplitude-modulation characteristics, and its evolutionary power spectral density (EPSD) varies with time [Kareem et al., 2019]; stationary power spectra and coherence functions therefore cannot be transferred without examination. Transient aerodynamic effects also change with pulse rise time and geometric configuration; the experimental results summarized in the review are not consistent, and the available evidence cannot provide universally applicable transient aerodynamic coefficients [Solari, 2020].

Under mixed wind climate conditions, including thunderstorm and non-thunderstorm events in the same extreme-value analysis sample may mask their different tail behavior and structural effects [Twisdale and Vickery, 1992]. The necessity of mechanism separation has been widely recognized, and corresponding probabilistic frameworks include separated marginal distributions, mixture models and Monte Carlo storm simulation methods [Aboshosha et al., 2020, Severe-thunderstorm stochastic model]. During the development of the non-hurricane hazard maps in ASCE 7-22, station observations were classified into thunderstorm and non-thunderstorm events, and peaks-over-threshold (POT) methods were applied to model each category separately before spatial smoothing [American Society of Civil Engineers, 2022] (Commentary C26.5.1, printed pp. 863-865). The standard introduces an independent tornado load architecture in Chapter 32, including dedicated coefficients V_T, K_{dT}, K_{zTor} and G_T, but these coefficients are derived for the tornado mechanism specifically and are non-transferable to downbursts [American Society of Civil Engineers, 2022] (Eq. 32.10-1.SI, printed p. 357).

Within the examined scope of ASCE 7-22 Chapters 26 through 31, the related Commentary, and the Chapter 32 tornado comparison, no downburst-specific transient wind-field-to-load response procedure has been identified [American Society of Civil Engineers, 2022]. This is a bounded review-scope statement, not a universal absence claim across all ASCE documents. How to translate the nonstationary hazard, spatially varying vertical profiles, finite-duration pulse dynamics and mechanism-dependent aerodynamic transfer of transient thunderstorm wind fields into structural design load formulations constitutes a critical interface gap in current wind engineering.

Several response-translation methods have been developed around this interface gap. The gust-front factor (GFF) proposed by Kwon and Kareem [Kwon and Kareem, 2009] pairs the ASCE 7 baseline with transient wind-field effects through a height-varying kinematic coefficient and a scalar response ratio. The subsequent generalized gust-front factor (G-GFF) [Kwon and Kareem, 2013] uses the gust-front static-mean response as denominator, removing the dependence on a specific code. GFF and G-GFF are two alternative bookkeeping containers for the same response information, and their comparison requires denominator alignment; they are not independent amplification factors that may be multiplied [Kareem et al., 2019, Kwon and Kareem, 2009, 2013]. Solari and coworkers developed the thunderstorm response spectrum technique (TRS) [Solari, 2016, Solari and Gaetano, 2015], expressing the structural peak response as the product of the peak static load and an equivalent response spectrum. Equivalent static wind loads (ESWL) start from a specific dynamic target effect and seek a load pattern that reproduces that effect in a static analysis [Chen and Kareem, 2001, 2005b, Denoël, 2024, Patruno et al., 2017b, Sun et al., 2023]. These methods each adopt different denominators, baseline loads, target responses and target statistics, and require definition alignment before comparison. Using the mechanism-conditioned hazard-to-load architecture as an organizing framework, the following sections systematically review the theoretical foundations, numerical characteristics, applicability boundaries and standard-interface conditions of these response-translation methods, and integrate GFF and G-GFF as candidate response-translation frameworks in the sections authorized for this integration.

2 Mechanism-conditioned hazard-to-load architecture

The complete path from transient thunderstorm wind fields to structural design loads can be organized as a compact mechanism-conditioned transformation chain [Kareem et al., 2019]. Let k denote the storm mechanism (e.g., downburst, gust front), a the structural type, r the target response quantity (e.g., base shear, overturning moment, displacement), and T the return period; the chain can then be written as

$$V_k(T) \longrightarrow u_k(z,t) = \overline{U}_k(z,t) + u'_k(z,t) \longrightarrow p_k(\boldsymbol{x},t) \longrightarrow E_{k,a,r}. \tag{1}$$

Here $V_k(T)$ is the mechanism-conditioned hazard wind-speed intensity, carrying an explicit reference height and averaging-duration definition and containing no structural response amplification. $u_k(z,t)$ is the wind-speed field varying with height z and time t, $\overline{U}_k(z,t)$ is its slowly varying mean component, and $u'_k(z,t)$ is the nonstationary fluctuating component. $p_k(\boldsymbol{x},t)$ is the surface pressure field at position $\boldsymbol{x}$ on the building surface at time t, with units of pressure (Pa); the conversion from surface pressure to forces, moments, generalized loads and structural response is accomplished through area integration and structural transfer, and p_k itself is not ambiguously described as a mixed force-pressure quantity. $E_{k,a,r}$ is the designated response effect or extreme-value statistic, whose units depend on the definition of the response quantity r.

The time-varying mean can be further factored as $\overline{U}_k(z,t) = V_k(T)\,P_k(z)\,A_k(t)$, where $P_k(z)$ is the normalized vertical profile function and $A_k(t)$ is the normalized temporal modulation function [Kareem et al., 2019] (Eq. 2, PDF p. 2). This factorization may only be used after the reference height, averaging duration and normalization conventions of P_k and A_k have been stated. This study does not imply that PETW currently provides a final $V_k(T)$.

In the expected engineering endpoints of PETW, the layers of the chain above correspond to identifiable evidence and computation sources. The Events layer will be used to distinguish downbursts, gust fronts and other severe convective mechanisms, and will provide the basis for event separation and occurrence-rate modeling in mixed wind climates. Extreme-value analysis (EVA) will determine return-period-conditioned mechanism-specific hazard inputs $V_k(T)$ with explicit reference heights and averaging durations on top of the mechanism classification. WRF will be used to describe the parent convective system and environmental conditions, and CM1 to resolve the downdraft, cold pool spreading, gust-front translation and the spatiotemporal evolution of the near-surface wind field. CFD or wind-tunnel testing will be used to establish and verify the aerodynamic transfer from nonstationary wind fields to building-scale surface pressure fields $p_k(\boldsymbol{x},t)$, resultant forces and force-moment time histories, and structural response analysis will compute the nonstationary response $E_{k,a,r}$ within the time-frequency framework of the generalized wind-loading chain [Kareem et al., 2019]. This narrative defines expected endpoints, not a completed PETW computation chain. Current PETW evidence cannot numerically close this complete chain: the downstream interfaces from WindField to EVA, structural response and DesignLoad have not received completed data packages, and $V_k(T)$, pressure-driven structural response, GFF, G-GFF and design load coefficients have not received final numerical values.

$V_k(T)$ stays at the hazard level and contains no structural response amplification. Embedding structural dynamics effects in the wind-speed parameter would conflate the independent hazard and response levels, leading to double counting [American Society of Civil Engineers, 2022] (Sec. 26.11.7, printed p. 279). ASCE 7-22 in Sec. 26.11.7 explicitly prohibits separately determining G where combined coefficients such as (GC_p) or (GC_{pi}) are provided, constituting a code precedent against double counting. In the transient wind-load context, if GFF, G-GFF or TRS already provides a complete mapping from static load to peak response, an additional gust factor or dynamic amplification coefficient containing the same response physics should not be stacked on top.

Classification uncertainty is the starting point of the chain. Correctly classifying an observed event in a mixed climate as a downburst, gust front, tornado or synoptic gust directly affects

the statistical modeling of all subsequent levels [Severe-thunderstorm stochastic model, Twisdale and Vickery, 1992]. The temporal resolution and spatial density of conventional weather stations are often insufficient to reliably identify local downbursts [Solari, 2020]. Mechanism-conditioned extreme-value analysis requires independent modeling of each mechanism branch after separation, followed by probabilistic recombination to obtain mixed return levels. Directly superposing mechanism-conditioned hazard onto ASCE map hazard without storm separation and probabilistic recombination is not permitted.

Every layer of transformation in Eq. (1) requires definition consistency across the dimensions of mechanism, height, averaging duration, reference state, aerodynamic assumptions, structural type a, response index r, target statistic and validation domain. The numerator and denominator of the response ratio must use the same structure, response quantity, response location or generalized coordinate, hazard level, averaging basis, aerodynamic model class and extreme-value definition [Kwon and Kareem, 2009, 2013].

Figure 1 illustrates this mechanism-conditioned hazard-to-load architecture and the definition-consistency requirements across its layers. The transformation at each layer corresponds to identifiable evidence sources within the examined corpus, and the independent evidence needed for calibration remains incomplete within the current review scope.

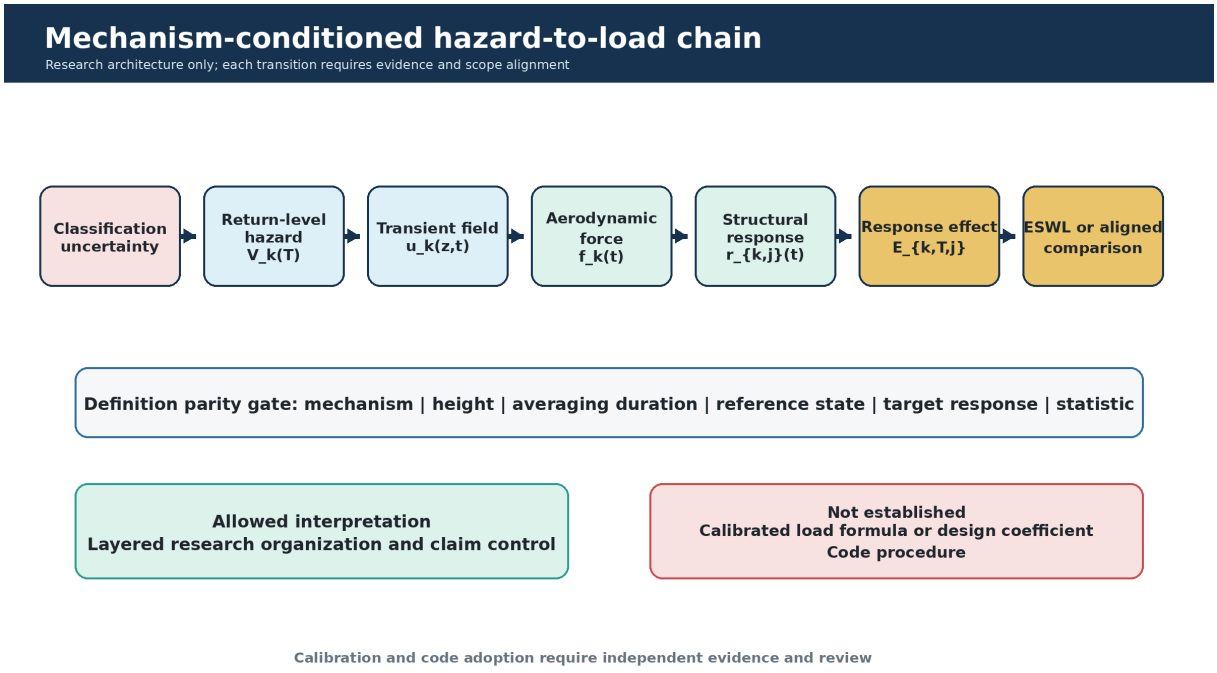


Figure 1: Mechanism-conditioned hazard-to-load architecture and definition-consistency requirements across layers.

3 Transient wind-field and aerodynamic-transfer representations

The downburst wind-speed field is commonly decomposed into a slowly varying mean component and a nonstationary fluctuating component: $U_{G\text{-}F}(z,t) = V_{G\text{-}F}(z,t) + u_{G\text{-}F}(z,t)$ [Kwon and Kareem, 2009] (Eq. 1, PDF p. 2). In the unified notation of this study, this corresponds to $u_k(z,t) = \overline{U}_k(z,t) + u'_k(z,t)$ in Eq. (1), where the subscript k identifies the mechanism. The mean component is further separated into the product of a spatial profile function and a temporal function: $\bar{U} = \bar{U}_0\,\bar{U}_H(z)\,\bar{U}_T(t)$ [Kareem et al., 2019] (Eq. 2, PDF p. 2). The reference

averaging period varies by event type; some downburst studies adopt event-specific moving-average windows, while Solari uses a 30 s moving average for the decomposition [Solari, 2016] (Eqs. 1-2, PDF p. 3).

Regarding the vertical profile, several analytical models are used to describe the nose-shaped characteristics of downbursts. The Vicroy model [Kwon and Kareem, 2009] (Eq. 4, PDF p. 3) is parameterized by $V_{\max}$ and $z_{\max}$; the Oseguera-Bowles model and the Wood model provide alternative parameterizations [Kwon and Kareem, 2019] (Eqs. 5-7, PDF p. 2), and significant differences exist among the three. Two criteria for matching these profiles to ASCE exposure categories (Criterion 1: equal speed at 10 m; Criterion 2: $V_{\max}$ equals the gradient-height gust) give numerically different load distributions [Kwon and Kareem, 2009] (Eqs. 6-9, PDF pp. 4-5), and this terrain dependence is hypothetical and has not been generally validated.

The simplified model for the temporal pulse commonly adopts a half-sine form $V_{G\text{-}F}(t) = \sin(\pi t/t_d)$ [Kwon and Kareem, 2009] (Eq. 3, PDF p. 3), where t_d is the pulse duration. This is a simplified envelope and does not represent an empirically universal waveform. The ratio of pulse duration to the first structural period t_d/T_1 is the key parameter controlling dynamic amplification; under the half-sine model, amplification reaches a maximum at $t_d/T_1 = 1$ [Kwon and Kareem, 2009] (Table 2) and approaches 1 for long pulses ($t_d \gg T_1$). Under long-pulse conditions, Kwon and Kareem derived the equivalent time $T_{eq} \approx 0.37\, t_d$ [Kwon and Kareem, 2019] (Eq. 23, PDF p. 7); this result is conditioned on the model adopted and on the long-pulse regime.

The fluctuating component is commonly treated as an amplitude-modulated stationary Gaussian process $u' = \sigma_v(z,t)\,\tilde{u}'(z,t)$ [Solari, 2016] (Eqs. 1-2, PDF p. 3), with evolutionary power spectral density $S_{uu}(f,t) = |A(f,t)|^2\, S_{\tilde{u}\tilde{u}}(f)$ [Kareem et al., 2019] (Eq. 7, PDF p. 3). Coherence functions at the current stage mostly adopt a time-invariant Davenport-type form [Kwon and Kareem, 2009, Solari, 2016], which is a provisional simplification. Wavelet transforms provide an alternative time-frequency representation [Kareem et al., 2019] (Eqs. 22-25). Observational data on downburst turbulence and coherence remain scarce, and statistical characteristics obtained from limited and geographically concentrated multi-event records [Solari and Gaetano, 2015] cannot automatically be generalized to all downburst events.

Regarding aerodynamic transfer, the conversion from transient wind fields to surface pressure $p_k(\boldsymbol{x}, t)$ admits two mutually exclusive evidence entries. In the simplified quasi-steady factor model, the transient aerodynamic departure is represented by the factor $I_4 = C_{D,k}/C_D$ [Kwon and Kareem, 2009] (Eqs. 17-19, PDF p. 6), where $C_{D,k}$ is the drag coefficient under mechanism k and C_D is the drag coefficient under stationary ABL conditions; simultaneously, the spectral and spatial coherence changes induced by the mechanism enter the nonstationary turbulence response modeling of I_3. If CFD or wind-tunnel testing provides a complete pressure time history $p_k(\boldsymbol{x}, t)$ incorporating transient aerodynamics, spatial correlation and directional evolution, this information should enter the generalized wind-loading chain [Kareem et al., 2019] directly, and the response $E_{k,a,r}$ should be computed through area integration and structural transfer, with GFF subsequently derived as a matched response ratio; in this case, the complete pressure-driven response must not be multiplied again by I_4, as doing so would double-count the aerodynamic effect. The same aerodynamic effect may appear at only one entry [Kareem et al., 2019, Kwon and Kareem, 2009].

Regarding site-specific hazard quantification, Monte Carlo storm simulation provides a mechanism-separated probabilistic framework. Aboshosha et al. developed a thunderstorm event detection and fitting workflow based on historical records for the Lubbock station [Aboshosha et al., 2020]; their event selection uses a gust factor ratio criterion $G_v/G_{vsy} > 1.25$, a one-hour separation window, and a ten-minute target resolution, and fits the event occurrence rate $\lambda = 4.76$ events/year with acceptance fit $R^2 > 0.8$ (mean $R^2 = 0.92$); these workflow parameters appear on PDF pp. 13-16. The workflow reports a 1000-year thunderstorm simulation value of about 58 m/s, a historical synoptic estimate of about 42 m/s, and a 700-year Monte

Carlo estimate of 56.8 m/s (PDF pp. 16-17). However, the source contains an unresolved internal conflict on the 700-year ASCE 7-16 comparison value: Figure 15 is labeled 47.0 m/s, while the page 17 text reports 49.0 m/s and gives an 11.5% comparison value [Aboshosha et al., 2020]. Both branches are retained without selecting either one. All values are outputs of the Lubbock-specific case, conditioned on station history, event classification, fitted outflow model and tail extrapolation, and cannot serve as transferable wind-speed parameters or ASCE 7-22 comparisons.

Regional-scale WRF simulations also face internal consistency challenges. Li et al. used WRF to simulate the extreme wind climate of the Arabian Peninsula [Li et al., 2021], employing convective instability proxies ($\mathrm{LI}_{\mathrm{crit}} < -2$ and $\mathrm{CAPE}_{\mathrm{crit}} >$ mean +0.1 standard deviation, PDF p. 8, Eq. 1) for thunderstorm identification and producing 50 to 3000 year return-period 3 s gust maps (PDF p. 14). The source contains an unresolved internal conflict: Table 1 specifies the modeled period as 1988-2017 (CFSR 1988-1999, GFS/FNL 2000-2017), while the body text on PDF pp. 3-4 states 1998-2017. Table 1 is the controlling configuration branch [Li et al., 2021]. The conflict is retained without normalization. These thresholds and periods are source-specific proxy indicators and do not constitute direct proof of downbursts or transferable design parameters.

Reanalysis-driven cyclone climatology and thunderstorm-related wind-field observations provide additional mechanism-identification evidence. Hourngir, in a 142-page doctoral thesis, presents CM1 configuration and LiDAR comparisons (Part 2 natural PDF pp. 3-9; Part 3 natural PDF pp. 5-8) and an ERA5-driven 1940-2022 single-algorithm cyclone climatology (Part 4 natural PDF pp. 1-8, 11-22) [Hourngir, 2025]. That source supports wind-field and hazard evidence only within its own research boundaries, and does not support structural-load validation, reliability calibration or standard-interface verification.

4 Structural response routes: GFF, G-GFF and TRS

In the process of translating transient wind-field effects into structural design loads, three principal response routes have been developed. Each route has a different denominator, baseline load, embedded physics and target statistic, and definition alignment must be performed before comparison. The relationship between GFF and G-GFF needs to be distinguished at two levels: in the reduced-route path, the literature decomposes the response ratio into the $I_1 I_2 I_3 I_4$ bookkeeping decomposition; in the complete pressure-driven generalized-chain path, the response $E_{k,a,r}$ is computed directly from the transient pressure field and the structural model, and GFF is subsequently derived as a matched response ratio. This study refers to the former as the reduced-route path and the latter as the direct-chain path; the two are mutually exclusive in the computational source of GFF.

4.1 Gust-front factor (GFF)

The original GFF proposed by Kwon and Kareem [Kwon and Kareem, 2009] uses the ASCE 7 baseline response as denominator and expresses the design force as $F_{\mathrm{Design}} = F_{\mathrm{ASCE7}}\, K_{z,G\text{-}F}\, G_{G\text{-}F}$ (Eq. 10, PDF p. 5). The scalar GFF is defined as the ratio of the gust-front maximum response to the ABL maximum response (Eq. 16), decomposed as $G_{G\text{-}F} = I_1\, I_2\, I_3\, I_4$ (Eq. 18, PDF p. 6). Here I_1 compares the static modal effect of the gust-front and ABL profiles (Eqs. 20-21, PDF p. 7), I_2 is the ratio of the peak response of the normalized half-sine mean load to its static response (Eq. 22), I_3 compares the nonstationary fluctuation response plus translation contribution to the conventional gust loading factor (Eq. 24, PDF p. 8), and I_4 is the ratio of transient to stationary aerodynamic drag coefficients. The height-varying kinematic coefficient $K_{z,G\text{-}F}$ is the normalized ratio of gust-front to ABL velocity-pressure profiles (Eqs. 25-26, PDF p. 9) and cannot be blindly multiplied by another profile correction.

In the unified notation of this study, the response-specific GFF ratio is written as

$$R^{\mathrm{GFF}}_{k,a,r} = \frac{E^{\max}_{k,a,r}}{E^{\max}_{\mathrm{base},a,r}} = I_1\, I_2\, I_3\, I_4, \tag{2}$$

where the denominator $E^{\max}_{\mathrm{base},a,r}$ must be obtained from a boundary-layer wind baseline that is to be fully recomputed under ASCE 7-22 procedures, and the numerator $E^{\max}_{k,a,r}$ is the peak response of the same structure a and same response quantity r under mechanism k. The current PETW case has not produced numerical values for this numerator or denominator. The $I_1 I_2 I_3 I_4$ decomposition in Eq. (2) is the bookkeeping decomposition of the reduced-route path provided in the literature [Kwon and Kareem, 2009] (Eq. 18), not a decomposition that has been shown to possess independent multiplicability in general. In the direct-chain path, $E^{\max}_{k,a,r}$ is computed directly from the complete pressure field $p_k(\boldsymbol{x}, t)$ and the structural model, and $R^{\mathrm{GFF}}_{k,a,r}$ is subsequently derived as a matched response ratio; it must not then be multiplied again by I_4 or have $I_1 I_2 I_3 I_4$ claimed as an independent re-decomposition.

The height-dependent kinematic coefficient paired with $R^{\mathrm{GFF}}_{k,a,r}$ is written as

$$K_{z,k}(z) = \frac{1}{I_1} \frac{V_k^2(z)}{(1/G_q)\, V^2_{\mathrm{base}}(z)}, \tag{3}$$

where G_q is the source-inherited velocity-pressure gust normalization whose role is to align the averaging-duration basis of the baseline wind speed so that the velocity pressure in the denominator is consistent with the averaging-duration convention of the original GFF derivation. G_q is not an additional ASCE 7-22 load factor; this study assigns no numerical value to it, and its exact implementation in ASCE 7-22 remains open until the baseline and averaging-duration conversion have been rederived. $V_k(z)$ is the wind speed of mechanism k at height z, and $V_{\mathrm{base}}(z)$ is the ASCE 7-22 baseline wind speed at the same height.

The corresponding candidate response-matched GFF equivalent load distribution can be expressed as

$$F^{\mathrm{GFF}}_{k,a,r}(z) = F_{\text{base,7-22},a,r}(z)\, K_{z,k}(z)\, R^{\mathrm{GFF}}_{k,a,r}, \tag{4}$$

where $F_{\text{base,7-22},a,r}(z)$ is the response-matched baseline equivalent load distribution that is to be fully recomputed under ASCE 7-22 procedures, not an arbitrary pressure equation. The ASCE 7-22 baseline exposure coefficient K_z is retained inside this baseline; $K_{z,k}(z)$ is a normalized transformation relative to that baseline, not an exposure coefficient applied again outside the baseline. $F_{\text{base,7-22},a,r}(z)$ and $F^{\mathrm{GFF}}_{k,a,r}(z)$ must use the same discretization and units: when expressed along the height, line loads (N/m) may be used; in a discrete structural model, nodal forces (N) may be used.

In the paired definition of Eq. (4), the $1/I_1$ contained in $K_{z,k}$ cancels the I_1 in $R^{\mathrm{GFF}}_{k,a,r}$, so that the net transformation relationship is

$$K_{z,k}(z)\, R^{\mathrm{GFF}}_{k,a,r} = \frac{V_k^2(z)}{(1/G_q)\, V^2_{\mathrm{base}}(z)}\, I_2\, I_3\, I_4. \tag{5}$$

This form separates the local velocity-pressure variation of the mechanism wind profile from the pulse dynamic factor I_2, the nonstationary turbulence response factor I_3 and the transient aerodynamic factor I_4, while ensuring that the kinematic effect I_1 is counted only once in the paired usage. It follows that multiplying $K_{z,k}$ by I_1 outside the pair is prohibited: the form $K_{z,k} \cdot I_1 \cdot I_2 \cdot I_3 \cdot I_4$ produces double counting of I_1.

For an aligned ABL configuration, when all mechanism-specific departures are eliminated, the decomposition in the source literature reaches the joint return limit: $R^{\mathrm{GFF}}_{k,a,r} \to 1$, $K_{z,k}(z) \to 1$, $F^{\mathrm{GFF}}_{k,a,r}(z) \to F_{\text{base,7-22},a,r}(z)$ [Kwon and Kareem, 2009] (PDF pp. 9-10). This joint return is a construction property of the reduced-route path; this study does not imply that every intermediate factor in the full-chain path can be independently observed or independently calibrated.

Because the original GFF algebra was derived on the basis of an earlier ASCE edition, in which the K_d placement and the K_z coefficient have both changed in ASCE 7-22 (see the standard-interface section below), the strict preservation of the GFF ratio structure in the candidate expression Eq. (4) after the 7-22 term rearrangement has not yet been rederived and verified, constituting an open technical requirement.

A parametric study on three idealized buildings, two ASCE exposures and four pulse durations shows that profile selection (Criterion 1 versus Criterion 2) dominates many load differences [Kwon and Kareem, 2009] (PDF pp. 9-11). Near $z_{\max}$, gust-front ESWL may exceed ABL loads even when $I_2\, I_3\, I_4 \leq 1$, primarily because of the kinematic effect of the nose-shaped profile. These numerical factors are model-dependent and case-dependent and do not possess generality.

4.2 Generalized gust-front factor (G-GFF)

Kwon and Kareem define G-GFF as the ratio of the gust-front maximum response to the gust-front static-mean response [Kwon and Kareem, 2013] (Eq. 8, PDF p. 3). In the unified notation of this study, the response-specific G-GFF is written as

$$G_{G,k,a,r} = \frac{E^{\max}_{k,a,r}}{\overline{E}_{\mathrm{st},k,a,r}} = I_2\,(I_3\, G_{\mathrm{GLF}})\,, \tag{6}$$

where the denominator $\overline{E}_{\mathrm{st},k,a,r}$ is the static response of the time-varying mean load under mechanism k, decoupled from any code baseline. G_{GLF} is the conventional gust loading factor taken from the target code and is taken once in the G-GFF definition. The profile factor I_1 and the drag ratio I_4 lie outside the generalized dynamic factor; when $C_{D,k}$ already appears explicitly in the self-contained G-GFF load expression, its departure must not be repeated through I_4.

For ABL wind, because $I_2 \to 1$ and $I_3 \to 1$, G-GFF reduces to the conventional gust loading factor G_{GLF} rather than to unity [Kwon and Kareem, 2013] (Eq. 9), which differs from the behavior of GFF reducing to 1 for boundary-layer wind. This distinction reflects the denominator difference between the two containers: GFF uses the ABL peak response as denominator, while G-GFF uses the mechanism-wind static-mean response as denominator.

The self-contained G-GFF equivalent load form is written as

$$F^{\text{G-GFF}}_{k,a,r}(z) = \tfrac{1}{2}\rho\, A(z)\, V_k^2(z)\, C_{D,k}(z)\, G_{G,k,a,r}, \tag{7}$$

where ρ is the air density, $V_k(z)$ is the wind speed of mechanism k at height z, $C_{D,k}(z)$ is the drag coefficient under the mechanism condition, and $G_{G,k,a,r}$ is the response-specific generalized factor defined by Eq. (6). $A(z)$ is the projected windward area: when $F(z)$ is a line load along height (units N/m), $A(z)$ is the windward projected area per unit height (m^2/m); when $F(z)$ is a discrete nodal force (units N), $A(z)$ is the windward projected area of the corresponding height interval (m^2).

Eq. (7) is code-independent at the aerodynamic and dynamic computation levels, but it is not a complete ASCE design load equation. The directionality factor K_d, the topographic factor K_{zt}, the ground elevation factor K_e, pressure coefficients, internal pressure, enclosure and opening classification, load cases, load combinations, the minimum load floor and reliability calibration must all be provided separately at the interface layer. $C_{D,k}(z)$ and G_{GLF} are each taken once from a single source and must not be repeated in other factors [Kwon and Kareem, 2013] (PDF pp. 3-4).

A frequency-domain and time-domain comparison for a 200 m building with a 5 s period shows that root-mean-square displacement and velocity achieve reasonable agreement under long pulses ($t_d = 200$ s) [Kwon and Kareem, 2013] (PDF pp. 5-6). The extreme-value response baseline from 100 000 time-domain simulations shows that the frequency-domain peak estimate is conservative across the four approximations tested, but its effect on I_3/G-GFF is small [Kwon

and Kareem, 2013] (Table 2). When the algebraic inputs of the original GFF and G-GFF are aligned, ASCE and Eurocode cases yield the same ESWL [Kwon and Kareem, 2013] (Table 3, PDF p. 8); this is an internal formulation verification guaranteed by aligned inputs, aerodynamic coefficients and structural models, and does not constitute broad external validation or a proof of universal equivalence.

4.3 Thunderstorm response spectrum (TRS)

The TRS technique developed by Solari and coworkers [Solari, 2016, Solari and Gaetano, 2015] converts the thunderstorm wind field and structural system into response spectra and an equivalent static formulation. The equivalent wind spectrum technique (EWST) replaces the spatial turbulence field with an equivalent fully coherent process filtered for coherence, so as to reproduce aerodynamic admittance effects [Solari, 2016] (Eq. 30, PDF p. 5). The equivalent response spectrum (ERS) is defined as $S_{d,eq} = d_{eq,\max}$ [Solari, 2016] (Eq. 48, PDF p. 6); multiplied by the peak static load, it yields the maximum response and ESWL [Solari, 2016] (Eqs. 49-50).

Within the scope of the linear oscillator and equivalent wind model adopted, full coherence gives $S_{d,eq} = S_d$, and full incoherence gives the basic response spectrum S_{db}, so that $S_{db} \leq S_{d,eq} \leq S_d$ [Solari and Gaetano, 2015] (Eqs. 42-45, PDF p. 9). These bounds are analytical model bounds that apply only to the linear response-spectrum/EWST formulation adopted and its practical response range; they cannot be generalized into universal physical laws for arbitrary wind fields, structures or nonlinear responses.

The empirical basis of TRS comes from a thunderstorm record set at the three Italian ports of Genova, La Spezia and Livorno [Solari and Gaetano, 2015], with a 30 s moving average used to decompose mean and fluctuation (PDF pp. 4-5). Individual spectra exhibit significant inter-event dispersion, while mean differences among ports are smaller [Solari, 2016]. The antenna mast case uses a preliminary 50-year $\hat{v} = 43.85$ m/s at 13 m height, a first frequency of 0.92 Hz and damping of 0.005 [Solari and Gaetano, 2015] (PDF p. 11). Three parameterizations give mean ERS values of 1.15 to 1.20 and mean-plus-one-standard-deviation values of 1.36 to 1.38 [Solari, 2016] (Tables 3-4). A second mast case reports mean ERS 1.20 and coefficient of variation 0.14 [Solari and Gaetano, 2015] (PDF p. 11). In the arc-lamp case, a mean spectrum of 1.70 and a mean-plus-one-standard-deviation of 2.01 yield different ESWL [Solari and Gaetano, 2015] (PDF p. 6). The choice of design spectrum (mean, mean plus standard deviation, or exceedance probability) has not been determined; these are all case results, not general factors.

The source contains an unresolved internal conflict [Solari and Gaetano, 2015]: the mast case on PDF p. 11 gives mean ERS 1.20 and coefficient of variation 0.14, whose direct product $1.20(1 + 0.14) = 1.368$ and subsequent calculations use about 1.37; another branch reports 1.99, but the current extraction reports no stable page locator for that branch. Both numerical branches are retained; 1.99 does not replace the subsequent calculation branch, and about 1.37 is not extrapolated as a general factor.

4.4 Aligned comparison of GFF, G-GFF and TRS

GFF, G-GFF and TRS are alternative bookkeeping containers within the same response-translation family [Kwon and Kareem, 2019, Mengistu and Repetto, 2024] and are not independent factors that may be multiplied. GFF uses the ABL peak response as denominator and reduces to 1 for boundary-layer wind; G-GFF uses the gust-front static-mean response as denominator and reduces to G_{GLF} for boundary-layer wind; TRS uses the peak static load as denominator. The three have different normalization baselines, and direct numerical comparison requires denominator alignment first. $R^{\mathrm{GFF}}_{k,a,r}$, the total response multiplier M, GFF, G-GFF, TRS and ESWL belong to the same family; multiplying any two of these as independent gains would double-count the response amplification.

The consistency of GFF and G-GFF in expressing the same target load effect is conditional: it holds only when wind inputs, averaging basis, aerodynamic coefficients, structural model and response definition are fully aligned, as in the internal formulation comparison reported by Kwon and Kareem [2013] in Table 3. This is an internal paired verification and does not constitute external validation or universal equivalence.

Kwon and Kareem compared GFF/G-GFF with TRST through the normalized comparison parameter λ [Kwon and Kareem, 2019] (Eqs. 26-29) and reported inconsistencies of less than 10% in specific aligned cases [Kwon and Kareem, 2019] (Table 4). However, that source notes that the earlier dimensionless TRST comparison procedure contained errors and has been replaced by a dimensioned response-spectrum formulation. Some corrected results needed to reconstruct the complete comparison chain are not available in that source (not reported). The less-than-10% consistency is limited to specific aligned cases after matching profile, turbulence intensity, size factor, static load denominator and structural inputs, and does not constitute universal equivalence between GFF/G-GFF and TRST [Kwon and Kareem, 2019].

Mengistu and Repetto performed a direct comparison and limited full-scale validation of TRS and G-GFF [Mengistu and Repetto, 2024]. In the slender-structure frequency range of 0.1 to 1 Hz, the output ratio between the two methods is about 0.8 to 1.5, narrowing to 0.8 to 1.2 in a specific subset (PDF p. 12). TRS over-predicts peak displacement by about 12% to 41% for two measurement events (PDF p. 16). That validation involves only two events and a single 16.6 m lamp post (PDF pp. 13-17) and is insufficient to establish an overall bias; consistency between methods may still mask compensating assumptions.

Figure 2 shows the denominator and response-route comparison for the GFF, G-GFF and TRS paths.

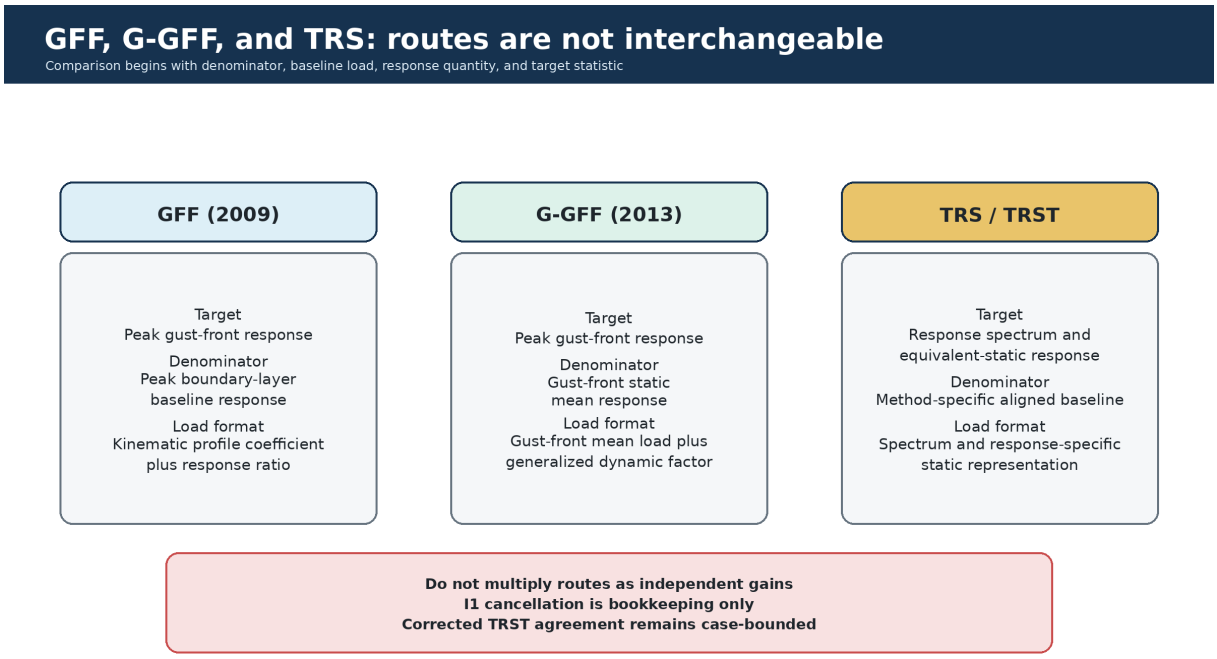


Figure 2: GFF, G-GFF and TRS response routes and denominator definitions.

5 Response-specific equivalent static wind loads

Equivalent static wind load (ESWL) is a response-translation technique whose objective is to find a set of static load patterns such that the response produced by that pattern matches

the target extreme-value effect of a dynamic analysis [Chen and Kareem, 2001, 2005b, Denoël, 2024, Park et al., 2019, Patruno et al., 2017b, Sun et al., 2023]. ESWL is not independent hazard evidence and not an additional load multiplier; it is a downstream static representation of an already-determined dynamic response, belonging to the same response-translation family as GFF, G-GFF and TRS.

The development of ESWL methods has progressed from single background/resonant decomposition to comprehensive modal coupling and envelope reconstruction. Chen and Kareem [Chen and Kareem, 2001] decomposed bridge buffeting response into background and resonant components and constructed static load patterns targeting each component separately. For three-dimensional coupled-mode buildings, the same decomposition framework was extended to include modal coupling and torsional effects [Chen and Kareem, 2005b], projecting aerodynamic forces onto each modal coordinate using high-frequency force balance (HFFB) measurements [Chen and Kareem, 2005a].

When coverage of multiple response extremes is needed, envelope reconstruction becomes the core problem. Patruno et al. [Patruno et al., 2017b] proposed an efficient ESWL determination method that approximately covers all response extremes through optimization of a finite set of static load patterns [Patruno et al., 2017a]. Denoël [Denoël, 2024] systematically quantified the complexity of the envelope reconstruction problem, comparing several ESWL methods under different response quantities, simultaneity conditions and load correlations. A key finding of that study is that a single static load pattern cannot reproduce the extreme values of all structural response quantities.

Patruno et al. [Patruno et al., 2018] performed a numerical study on ESWL extraction in buffeting analysis, comparing the accuracy and robustness of different extraction strategies. Park et al. [Park et al., 2019] evaluated and compared ESWL with database-assisted design (DAD), in which DAD directly uses the extreme-value responses from time-history analysis without static load conversion. For tall buildings, ESWL is less accurate than DAD, but is easier to integrate into practical design workflows [Park et al., 2019]. Sun et al. [Sun et al., 2023] reviewed ESWL estimation methods and their limitations for long-span roof structures. Local wind pressure coefficient generation for low-rise buildings [Kumar and Stathopoulos, 2001] and field-measurement/wind-tunnel/code comparisons for low-slope gable roofs [Chavez et al., 2022] provide aerodynamic evidence at the C&C level, but ESWL evidence obtained under synoptic conditions cannot be directly transferred to transient thunderstorm loading.

In the transient downburst context, GFF and TRS frameworks each produce a corresponding ESWL interface [Kwon and Kareem, 2009, Solari, 2016], but its validity is limited by the along-wind model and modal assumptions adopted. If GFF, G-GFF or TRS already provides a complete mapping from static/mean load to peak response, ESWL should not be stacked on top of that response formulation as another factor, as doing so would produce double counting. ESWL is an output formulation aimed at specific response quantities, not an additional response gain.

When the response extends beyond the literature coverage of along-wind MWFRS, first linear mode and regular structures, a single scalar response ratio or factor cannot simultaneously reproduce multiple structural effects such as base shear, overturning moment and displacement; in such cases, switching from a scalar factor to a response-specific ESWL load distribution targeted at the particular response quantity r is required. Switching criteria include: the response involves crosswind, torsion or C&C local peaks; higher-mode contributions are not negligible; the load distribution is non-monotonic along the height such that a single scalar ratio cannot simultaneously reproduce multiple target effects; and CFD or wind-tunnel pressure-field spatial correlation structures depart significantly from Davenport coherence. Under these conditions, the ESWL distribution must be constructed through equilibrium-consistent, response-specific load cases, or through comparisons formed against the same target effect and reliability basis. A pointwise heightwise maximum envelope of two load distributions with different height pro-

files must not be taken without accounting for spatial correlation, force equilibrium and response specificity, unless a dedicated ESWL reconstruction has demonstrated that the operation is valid for the target effects under consideration.

Figure 3 illustrates the ESWL response-translation flow and its double-counting guards.

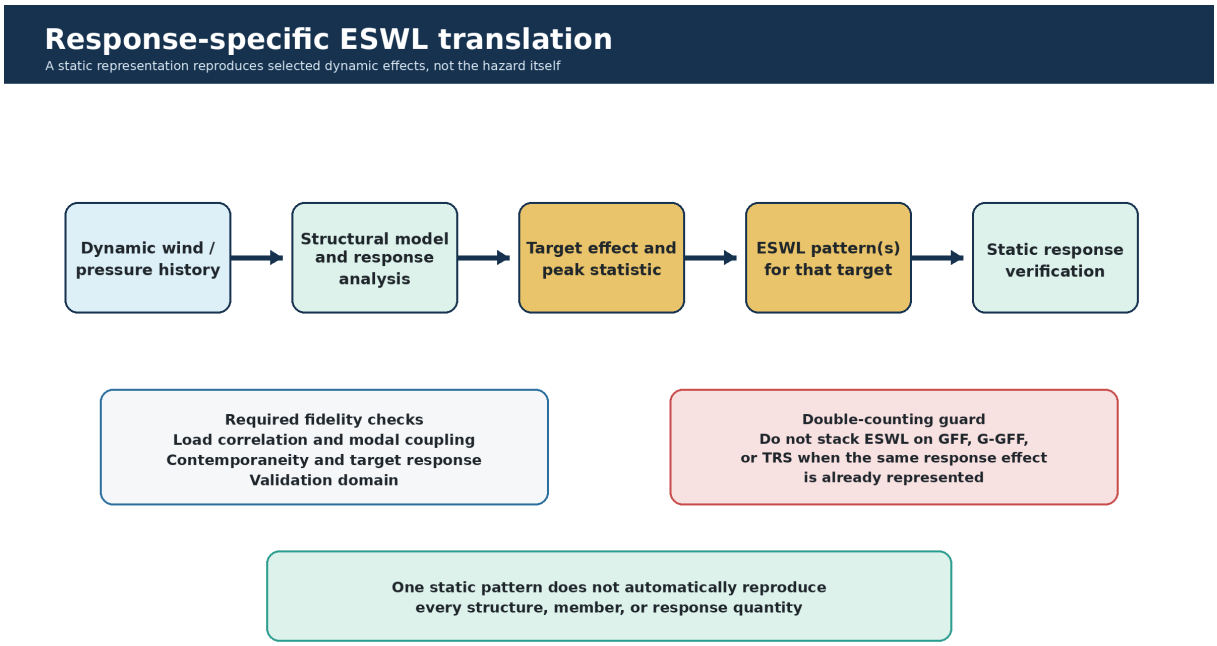


Figure 3: ESWL response-translation flow and double-counting guards.

6 Standard interface and edition-specific bookkeeping

Mapping transient thunderstorm wind-field effects onto structural design codes requires precise preservation of edition-specific term placement and coefficient definitions. ASCE 7-22 [American Society of Civil Engineers, 2022] and ASCE 7-16 [American Society of Civil Engineers, 2017] differ critically in the structure of the velocity-pressure formula.

ASCE 7-16 places the directionality factor K_d inside the velocity pressure (Eq. 26.10-1), and its exposure coefficient table uses a 2.01 cap. ASCE 7-22 removes K_d from q_z, placing it in downstream pressure or force equations (e.g., Eq. 27.3-1, $p = q\,K_d\,G\,C_p - q_i\,K_d\,(GC_{pi})$, printed p. 281), and the exposure coefficient K_z uses 2.41 and a recalibrated z_g [American Society of Civil Engineers, 2022] (Table 26.10-1, printed pp. 275-278). Mixing 2.01 with the 7-22 tables, or placing K_d inside the 7-22 q_z, would therefore produce edition-conflation errors [American Society of Civil Engineers, 2017].

In the mechanism-conditioned candidate interface, each coefficient and physical effect must follow a one-effect-one-owner allocation principle. The ASCE 7-22 baseline exposure coefficient K_z is retained inside the baseline load $F_{\text{base,7-22},a,r}(z)$, which is to be fully recomputed. $K_{z,k}(z)$ in Eq. (3) is a normalized kinematic transformation relative to that baseline, whose role in the paired GFF bookkeeping is to map the velocity-pressure variation of the mechanism wind profile onto the baseline; it is not an exposure coefficient applied again outside the baseline and cannot overlap with the K_z inside the baseline. The directionality factor K_d continues to reside in the downstream pressure or load equation of ASCE 7-22, controlled by the code layer rather than by the mechanism branch. The topographic factor K_{zt} is taken as 1 in the GFF family, with no terrain coverage; sites with terrain must have terrain handled separately by the ASCE layer,

and it must not be assumed by default that GFF results already contain terrain. The ground elevation factor K_e, pressure coefficients C_p and combined internal pressure coefficient (GC_{pi}), enclosure and opening classification, load cases and load combinations all remain at the code layer and are not replaced or duplicated by the mechanism translation.

In the candidate GFF interface, the mechanism branch changes the final velocity-pressure profile relationship and the dynamic response level, but this change must be accomplished through the pairing of a complete baseline with a relative transformation. The baseline K_z is retained in the denominator reference of $F_{\text{base,7-22},a,r}(z)$; $K_{z,k}$ and the I_1 in R^{GFF} together transform that baseline profile into the mechanism profile, not by deleting K_z or by applying a profile factor outside the baseline again. Dynamic response may be carried by the reduced representation $I_2 I_3$ or by the explicit response computation of the chain; the two cannot be counted simultaneously. Changing only the code gust-effect factor G/G_f without addressing the kinematic difference of the nose-shaped profile would omit the principal profile effect in the GFF literature. In the G-GFF self-contained path, the mechanism profile enters the mean load directly through $V_k^2(z)$, and the ASCE K_z chain does not serve as the generator of that aerodynamic load.

The wind load procedures of ASCE 7-22 cover different building types and structural systems through multiple chapters. Chapter 27 is the MWFRS directional procedure for all-heights regular buildings, with a built-in dynamic scope for along-wind gust amplification and resonance; crosswind response, vortex-induced vibration, galloping, flutter, wake buffeting or unusual shape/response require recognized literature or Chapter 31 [American Society of Civil Engineers, 2022] (Sec. 27.1, printed p. 281). Chapter 28 is the MWFRS envelope procedure for qualifying low-rise regular buildings. Chapter 30 covers C&C and is not interchangeable with the MWFRS chapters. Chapter 31 permits physical wind-tunnel procedures and verified, validated and independently peer-reviewed numerical procedures, but Sec. 31.4.4 imposes minimum-load controls on MWFRS and C&C loads relative to Chapters 27 through 30 [American Society of Civil Engineers, 2022] (printed pp. 351-352). Any mechanism-conditioned load produced via the Chapter 31 route must satisfy these minimum load floors, not replace them.

Chapter 32 creates an independent tornado load architecture for qualifying Risk Category III and IV structures [American Society of Civil Engineers, 2022] (Eq. 32.10-1.SI, printed p. 357), including dedicated mapped wind speed V_T, effective plan area, tornado directionality K_{dT}, tornado exposure coefficient K_{zTor}, velocity pressure $q_{zT} = 0.613\, K_{zTor}\, K_e\, V_T^2$, gust effect G_T, enclosure/internal pressure including atmospheric-pressure-change effects, vertical-wind external-pressure adjustments, and a bounded wind-tunnel route. The Chapter 26 exposure and terrain provisions do not simply carry over to Chapter 32. This modularity is an architectural precedent demonstrating that a code can establish an independent load branch for a specific mechanism, but tornado coefficients V_T, K_{dT}, K_{zTor}, G_T, (GC_{piT}) and K_{vT} are non-transferable to downbursts. Commentary C32.18 states that, due to the lack of consensus on testing conditions, methods and data analysis, tornado simulators do not currently constitute a general route to pressure/force coefficients [American Society of Civil Engineers, 2022] (printed pp. 941-942).

The candidate GFF interface expression Eq. (4) has not yet been rederived or calibrated as an ASCE 7-22 procedure. It is not a completed or approved ASCE 7-22 procedure. The original GFF algebra was developed on the basis of an earlier ASCE edition, in which K_d was inside q_z, the exposure coefficient was capped at 2.01 and the wind importance factor was retained; in ASCE 7-22, K_d has moved downstream, K_z is capped at 2.41 and the wind importance factor has been eliminated. After these term rearrangements, the strict preservation of the GFF ratio structure must be rederived and verified; that verification has not yet been completed.

International code comparisons provide context for understanding different G/G_f definitions. Zhou et al. [Zhou et al., 2002] performed a comparative study of along-wind load effects on tall buildings across major international codes, revealing differences among codes in gust factor definition, exposure coefficient and reference height. The methodological basis for wind-load

codification [Simiu and Stathopoulos, 1997] highlights the roles of bluff-body aerodynamics and climate databases in code development. The methodology for obtaining unified aerodynamic coefficients for ASCE 7 wind load provisions [Guo et al., 2025] illustrates the direction of current code evolution but does not address downburst-specific coefficients. The impact of climate change on extreme windstorms and structural design [Wu et al., 2026] is a frontier topic but does not support design conclusions or calibration within the current review scope.

Within the examined scope, ASCE 7-22 incorporates thunderstorm and non-thunderstorm station observations into the development of non-hurricane wind-speed maps but does not provide mechanism-specific downburst wind-field, aerodynamic or structural response procedures [American Society of Civil Engineers, 2022]. The existence of the tornado chapter demonstrates the feasibility of mechanism-specific modular design; the gap for a downburst interface is not a lack of attention but a lack of sufficient calibration evidence. Any attempt to map mechanism-conditioned transient hazard onto this baseline chain requires explicit definition of the placement of every coefficient, the physical effects embedded and the overlap boundaries relative to the existing ASCE factors.

Figure 4 illustrates the edition-specific bookkeeping differences between ASCE 7-16 and 7-22, the non-transferable tornado branch, and the open downburst calibration layer [American Society of Civil Engineers, 2017, 2022].

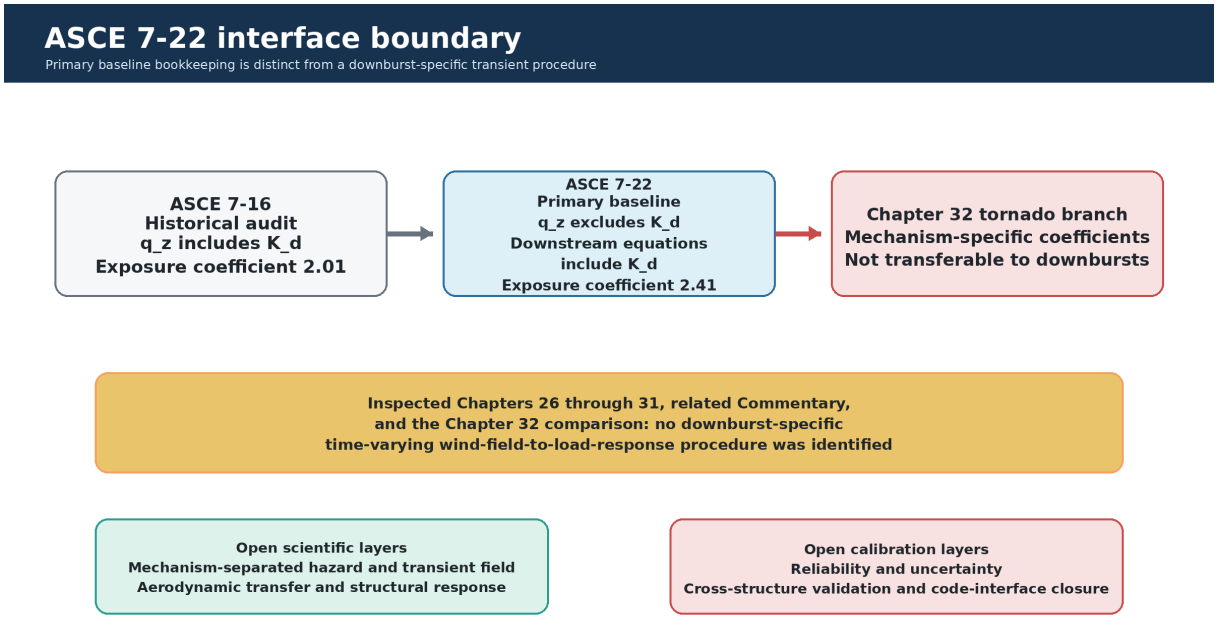


Figure 4: ASCE 7-16 versus 7-22 edition-specific bookkeeping differences, tornado branch and open downburst calibration layer.

7 Uncertainty, reliability and conservatism gaps

Translating transient thunderstorm wind-field effects into reliable design load formulations requires quantification of uncertainty at multiple levels: hazard (event occurrence rate, intensity distribution, return-period extrapolation), wind field (profile model, pulse duration distribution, turbulence characteristics), aerodynamic transfer (transient drag, pressure-field spatial distribution), structural response (modal parameters, damping, nonlinearity), model form (idealizations and departures of the analytical framework), and standard interface (physical effects embedded and not embedded in existing factors).

Kwon and Kareem performed a preliminary Monte Carlo uncertainty analysis of GFF/G-GFF [Kwon and Kareem, 2019] (Table 3), using six lognormal uncertain inputs with assumed coefficients of variation of 0.20 each for $V_{3\text{-}s}$, $z_{\max}$, t_d and c_1, 0.05 for frequency and 0.40 for damping. The 10 000 Monte Carlo simulations on the CAARC standard building produced preliminary load factors exceeding 3. Wind speed $V_{3\text{-}s}$ dominates the variability of the base bending moment, followed by profile height; turbulence intensity dominates the uncertainty of G-GFF. However, the authors of that study explicitly warn that these values primarily reflect assumed aleatory uncertainty distributions and coefficients of variation, omitting epistemic model uncertainty [Kwon and Kareem, 2019]. These preliminary load factors are assumption-driven source results and do not constitute code load factors or reliability calibration.

Life-cycle cost comparison of downbursts and tornadoes provides a cross-mechanism framework perspective [Le and Caracoglia, 2021], but the tornado branch in that analysis is non-transferable to downburst load coefficients. Differences between tornadoes and downbursts in spatial scale, duration, vertical profile and rotational characteristics make cross-mechanism coefficient transfer both unsupported and unsafe.

Within the scope of full-text literature and standards included in this review, no calibrated transient wind-load reliability factors, uncertainty factors, conservative defaults, equivalent wind-speed parameters or overall multipliers have been identified. This absence conclusion describes the evidence ceiling of the present corpus only and does not constitute a universal absence claim across all literature.

Computational wind engineering, machine-learning-assisted extreme-value extrapolation and climate-change-driven extreme windstorm evolution [Kareem, 2020, Wu et al., 2026] may provide research pathways for subsequent uncertainty quantification, but current evidence does not support design conclusions or calibration results.

8 PETW case study: upstream benchmark reconstruction and definition sensitivity

Moore et al. [Moore and Dawson, 2024] report a set of CM1 idealized downburst simulations and their near-surface wind-field diagnostics. The identity of the PETW case is supported by the configuration and selection logic in the authors' published script, but has not yet received direct confirmation from the article figure caption. The 7200 s reduced two-dimensional output trajectory and the 2220 s sparse three-dimensional branch maintain byte-identical common two-dimensional diagnostic fields at the 2220 s shared time step on a per-variable, per-tile basis. This result establishes computational and post-processing consistency under controlled output partitioning conditions; it does not indicate agreement with the article results and does not constitute exact reproduction, PETW Oklahoma Path C verification, wind-field validation or structural-load validation.

Decomposition invariance and bounded initial-condition tests further constrain the scope of definition sensitivity. Under the 5×8 and 6×8 decomposition schemes, the key wind-field variables at 38 common output times maintain byte-identical values. In the B and B-prime paired $t = 0$ test, after removing the explicit $z = 0$ m wind-profile row, 14 requested variables remain byte-identical; the lowest model level of the current grid is 50 m, and both configurations use the same 10 m to 73.48 m interpolation interval. The common-convention test reduces the 50 m initial vector wind speed from 6.1878911 m/s to 6.1156347 m/s, and the CM1 near-surface-layer diagnostic $s10$ from 1.9812088 m/s to 1.9580742 m/s, both with a change rate of −1.1677%; placing the source wind speed on the 10 m sounding row does not constrain $s10$ to equal that input value. The input sounding anchor value, the lowest model level vector wind and the near-surface-layer diagnostic must retain distinct variable identities; none of these is equivalent to a validated 10 m 3 s gust or $\hat{V}_{3s}$.

The controlled vertical-reference experiment isolates single-factor sensitivity. The PETW

candidate vertical-reference convention changes only 36 non-ground sounding heights, while the ground layer remains unchanged. The archived extractor regression test exactly reproduces the baseline values, thereby controlling for code drift. The candidate convention is a PETW-internal vertical-reference mapping and is not equivalent to the conversion method of Moore et al., because the available evidence cannot determine the vertical-coordinate treatment adopted by the authors.

The multi-metric results exhibit indicator-dependent directionality and cannot select an overall winner. At the official cross-section $j = 270$, the width at $\leq -2\,\mathrm{K}$ increases from 3.000 km to 3.800 km (digitized target about 5.431 km), and the minimum decreases from $-7.475\,\mathrm{K}$ to $-8.026\,\mathrm{K}$ (digitized target about $-8.136\,\mathrm{K}$), with the improvement direction consistent with the target; however, the corresponding area increases from $1.440\,\mathrm{km}^2$ to $1.530\,\mathrm{km}^2$ (digitized target about $1.404\,\mathrm{km}^2$), with area exceeding the target while width and minimum improve. The 0 to 5 km vertical profile RMSE degrades from 4.886 m/s to 6.181 m/s, and MAE from 3.636 m/s to 4.666 m/s. Footprint areas at thresholds above 20, 25 and 30 m/s show mixed directions: above 20 m/s area increases from $14.800\,\mathrm{km}^2$ to $18.730\,\mathrm{km}^2$ (digitized target about $13.446\,\mathrm{km}^2$), with increasing departure; above 25 m/s area decreases from $4.920\,\mathrm{km}^2$ to $4.750\,\mathrm{km}^2$ (digitized target about $6.739\,\mathrm{km}^2$), both under-predicting; above 30 m/s area increases from $0.660\,\mathrm{km}^2$ to $0.840\,\mathrm{km}^2$ (digitized target about $1.049\,\mathrm{km}^2$), with the gap narrowing. All article values are approximate raster values digitized from Figure 5 of Moore et al. [Moore and Dawson, 2024], subject to anti-aliasing, line width and color-scale mapping uncertainty, and are not the authors' original arrays. The overall effect of the candidate vertical reference is indicator-dependent: width and minimum improve while area exceeds and error degrades, and an overall calibration cannot be established.

The unresolved 2160 s versus 2220 s time semantics is a critical uncertainty in the current evidence. The authors' script selects file 000037 and labels Panel D 2160 s, while the completed sparse three-dimensional comparison uses 2220 s. Under the file-numbering convention adopted by this project, file 000001 corresponds to time zero, so file 000037 maps to 2160 s. This mapping remains a script-indicated hypothesis to be tested, because neither the authors' actual file list nor the article figure caption has been directly confirmed. The digitized curve peak from Figure 5 of Moore et al. [Moore and Dawson, 2024] lies at about 2160.96 s; the conditional proximity between the 2220 s fixed-point diagnostic and the raster shoulder marker may merely result from a sampling-time difference. The paired 2160 s uniform-reference and candidate-reference sparse three-dimensional diagnostic has not been submitted, and the existing 2220 s proximity therefore cannot be interpreted as calibration success.

The PETW case instantiates the definition, reproducibility and time-semantic gating described by the mechanism-conditioned chain of Eq. (1), but cannot provide numerical closure for that chain. The diagnostic results above can constrain downstream interfaces, but the WindField level has no completed mechanism-specific characterization data package, the EVA level has no return-level results, the structural response level has no response evidence driven by $p_k(\boldsymbol{x}, t)$ or corresponding load time histories, and the DesignLoad level has no GFF, G-GFF or approved coefficients. $V_k(T)$, pressure-driven structural response, GFF, G-GFF, reliability factors and design load coefficients have not received final numerical values. Variable identity, height reference, sampling time, spatial tiling, diagnostic definition and multi-metric acceptance criteria all need to be explicit fields in the data package. This study does not select GFF, G-GFF, TRS or ESWL as the PETW closure route, and does not place any CM1 diagnostic directly into the ASCE velocity-pressure chain.

9 Discussion

The three response routes (GFF, G-GFF and TRS) and ESWL each provide a different framework for mapping transient wind-field effects to structural design formulations; they belong to

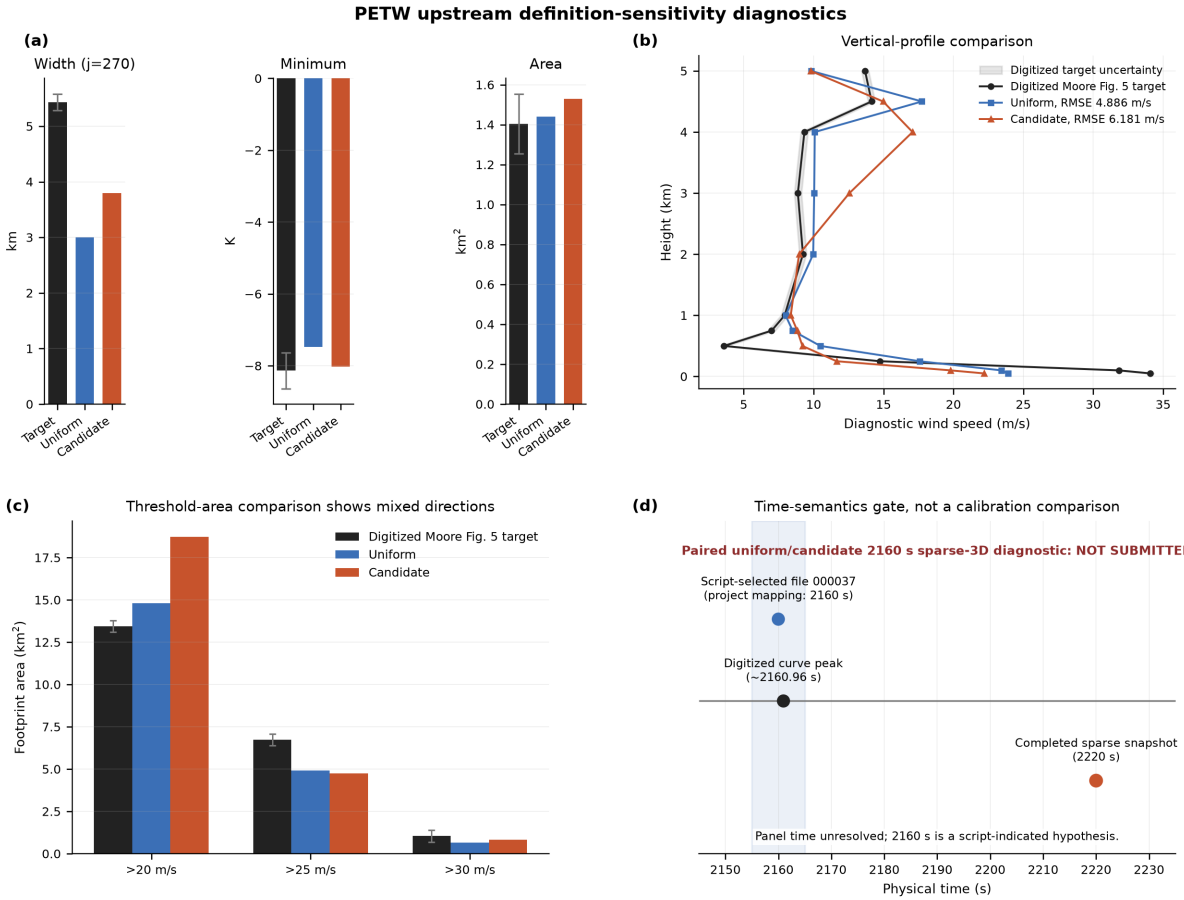


Figure 5: PETW upstream benchmark diagnostic summary. This figure is redrawn from machine-readable internal metrics and contains no pixels or cropped content from the Moore article. (a) Uniform-reference, candidate-reference and digitized-target comparison of cold-pool width, minimum value and area at the official cross-section $j = 270$; (b) 0 to 5 km vertical profile comparison, with uniform-reference and candidate-reference RMSE of 4.886 and 6.181 m/s respectively; (c) Footprint area comparison at thresholds above 20, 25 and 30 m/s. Digitized targets are extracted from the raster image of Figure 5 of Moore et al. [Moore and Dawson, 2024], subject to anti-aliasing, line width and color-scale mapping effects, with approximate uncertainties of width ±0.15 km, minimum value ±0.5 K, cross-section area ±0.15 km^2, profile wind speed ±0.25 m/s and footprint area ±0.35 km^2; these targets are not the authors' original arrays. (d) Time-semantics timeline, distinguishing the script-selected 2160 s, the digitized curve peak at about 2160.96 s, and the completed 2220 s sparse snapshot; the paired 2160 s diagnostic has not been submitted. This figure is used to show indicator dependence and the time-semantic gap, and does not constitute a calibration comparison.

the same response-translation family and are not independent gains that may be multiplied. Comparison among them reveals several key methodological issues.

The I_1 cancellation expressed in Eq. (5) is the core construction property of the GFF and $K_{z,k}$ paired bookkeeping. I_1 appears in the numerator of $R^{\mathrm{GFF}}_{k,a,r}$ as the height-integrated kinematic mean-load ratio in the first mode, while $1/I_1$ is embedded in normalized form in the definition of $K_{z,k}(z)$. The two cancel in the product, so that the net transformation relationship contains only the local height-by-height velocity-pressure ratio and the dynamic factors $I_2 I_3 I_4$. This cancellation prevents double counting of the kinematic effect and simultaneously means that, within a context where the pair already exists, I_1 must not be multiplied in separately outside $K_{z,k}$. In the G-GFF path, I_1 and I_4 lie outside the generalized dynamic factor, $C_{D,k}$ appears explicitly in Eq. (7) and is taken once, and the same drag ratio must not be counted again as I_4.

The single-entry rule for aerodynamic evidence is another double-counting constraint. In the simplified factor model, the transient aerodynamic departure enters $I_4 = C_{D,k}/C_D$ and the spectral and coherence model update of I_3. In the complete pressure-driven generalized-chain model, the surface pressure time history $p_k(\boldsymbol{x}, t)$ supplied by CFD or wind-tunnel testing already incorporates transient aerodynamics, spatial correlation and directional evolution and directly drives the structural response computation; GFF is then derived as a response ratio. The same aerodynamic effect cannot appear at both entries simultaneously; multiplying a complete pressure-driven response by I_4 again would double-count transient aerodynamics.

Denominator alignment is a prerequisite for valid comparison. GFF uses the ABL peak response as denominator and reduces to 1 for boundary-layer wind; G-GFF uses the gust-front static-mean response as denominator and reduces to G_{GLF} for boundary-layer wind; TRS uses the peak static load as denominator. Even when input wind fields and structural parameters are identical, a numerical comparison without denominator alignment lacks physical meaning. Kwon and Kareem achieved limited case-by-case comparison through explicit normalization, but the corrected dimensioned TRST results are partly unavailable and the complete comparison chain cannot be reconstructed [Kwon and Kareem, 2019] (Eqs. 26-29; Table 4). The agreement of GFF and G-GFF in Table 3 of Kwon and Kareem [2013] is premised on full alignment of wind inputs, aerodynamic coefficients, structural model and response definition; that agreement is an input-specific internal verification and does not constitute universal equivalence.

Response specificity pervades the entire framework. The subscript r in Eqs. (2) through (7) indicates that a response ratio calibrated for displacement cannot automatically represent base shear or overturning moment. A single $R^{\mathrm{GFF}}_{k,a,r}$ is acceptable only under assumptions where the modal shape and load pattern make the target effects proportional. When target effects are not proportional, higher modes are not negligible, or the load distribution is non-monotonic along the height, an independent $R^{\mathrm{GFF}}_{k,a,r}$ with a corresponding ESWL load pattern must be provided for each response-quantity family. Crosswind, torsion, C&C local peaks and internal-external pressure correlation therefore require treatment as separate branches, each needing dedicated aerodynamic and validation evidence.

The validation domain is not continuous across methods. The numerical evidence for GFF and G-GFF comes primarily from parameterized cases and limited time-domain benchmarks, using simplified assumptions such as half-sine pulses, Vicroy/Oseguera-Bowles/Wood profiles, Davenport spectra and time-invariant coherence functions [Kwon and Kareem, 2009, 2013]. TRS evidence comes from geographically concentrated multi-event records and mast or lamp post cases [Solari, 2016, Solari and Gaetano, 2015]. The full-scale validation by Mengistu and Repetto involves only two events and a single 16.6 m lamp post (PDF pp. 13-17) [Mengistu and Repetto, 2024]. These results cannot automatically be generalized to other structural types, events or climatic conditions.

The PETW case further demonstrates that, even when improvement is achieved on a single local diagnostic or cross-section metric, validation of a wind field or structural load cannot

be established when another metric degrades and sampling semantics remain unresolved. The bounded PETW case instantiates the definition, reproducibility and time-semantic gating of the chain described by Eq. (1), but cannot provide numerical values of $V_k(T)$, pressure-driven structural response, GFF, G-GFF or design load coefficients for that chain.

The future transition toward code-readable formulations requires meeting multiple preconditions: obtaining calibrated hazard inputs in representative climate samples, establishing reliable profile and turbulence parameterizations, achieving experimental or full-scale validation covering multiple structural types and response indicators, systematically quantifying model form uncertainty, and completing calibration within a target-reliability-index framework. The principles of mechanism separation and kinematic alignment provide research directions for this transition, but within the current review scope they do not constitute approved design quantities or candidate code provisions. Within the reviewed corpus, the tornado load architecture (ASCE 7-22 Chapter 32, printed p. 357) [American Society of Civil Engineers, 2022] and stochastic tornado trajectory models [Fan and Pang, 2019, Twisdale et al., 2023] demonstrate the feasibility of mechanism-specific modular design; however, the empirical basis and coefficients of the tornado branch are non-transferable to downbursts.

10 Concluding remarks

The mechanism-conditioned hazard-to-load architecture maintains the hazard intensity, spatiotemporal wind field, aerodynamic force transfer and structural response of transient thunderstorm winds as mutually connected but independently evidenced levels. This study integrates GFF and G-GFF as two candidate and mutually exclusive response-translation containers within this architecture. GFF, through the response-specific ratio $R^{\mathrm{GFF}}_{k,a,r}$ defined in Eq. (2) and the paired kinematic coefficient $K_{z,k}(z)$ defined in Eq. (3), expresses the mechanism-conditioned load as a multiplier on a baseline load that is to be fully recomputed under ASCE 7-22 procedures (Eq. 4), in which I_1 cancels in the paired usage (Eq. 5) and the pair jointly returns to the baseline for boundary-layer wind. G-GFF, through the generalized factor $G_{G,k,a,r}$ defined in Eq. (6) and the self-contained load form of Eq. (7), expresses the mechanism-conditioned load as an independent formulation decoupled from any specific code format, reducing to G_{GLF} for boundary-layer wind. TRS uses the peak static load as reference and compresses coherence and structural dynamics through the equivalent wind spectrum. The three have different denominators and embedded physics, so cross-method comparison must first align the reference state and verify double-counting boundaries. ESWL is a downstream static representation targeted at specific response effects and is not an additional factor that may be further multiplied with these routes. GFF, G-GFF, TRS, total response multipliers and ESWL belong to the same family and are not cumulative. The directionality factor placement and exposure coefficient differences between ASCE 7-22 and historical editions, the minimum-load controls of Chapter 31, and the boundary that Chapter 32 tornado branch coefficients are non-transferable to downbursts together delimit the bookkeeping conditions for any subsequent interface research [American Society of Civil Engineers, 2017, 2022, Chen and Kareem, 2001, Kwon and Kareem, 2009, 2013, Solari, 2016].

The available evidence still lacks cross-regional calibration of mechanism-conditioned hazard inputs, systematic full-scale validation across multiple structural types and response indicators, independent quantification of model form uncertainty, and a complete calibration consistent with target reliability. The examined sources also retain several unresolved internal conflicts, including inconsistencies between spectral-factor calculation branches, regional hazard comparison values, modeled period records, and the TRST dimensioned-correction chain; these conflicts cannot be eliminated through model voting or silent normalization [Aboshosha et al., 2020, Kwon and Kareem, 2019, Li et al., 2021, Solari and Gaetano, 2015]. The bounded PETW upstream benchmark case shows that the effect of the candidate vertical-reference convention is indicator-

dependent, the sampling time semantics remains unresolved, and the downstream interfaces from WindField to EVA, structural response and DesignLoad have not achieved numerical closure. GFF and G-GFF have been integrated into this study as candidate response-translation frameworks, but this integration is an expected research endpoint that still requires complete computation, experimental validation, reliability calibration and PI review before it can be confirmed. Until these evidence gaps are closed, GFF, G-GFF, TRS and ESWL should be maintained as comparable research methods rather than being promoted to closed design procedures, calibrated multipliers or formal code provisions.

Data and artifact availability

This manuscript is an internal PI review version. Source inventories, machine-readable figure data, generation code, claim ledgers and internal computation records are retained in the governed local work package, but local paths and Agent Updates cannot be cited as public literature. Prior to external submission, a supplementary data repository with DOI must be established, incorporating case definitions, run inventories, post-processing definitions, source data, figure generation code and quality assurance records; that repository has not yet been established.

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